Research Intake 2026

Day 23

ABSA & Emotion Recognition

A Beginner's Guide for Students

From Foundational Concepts to Research-Level Understanding

Gudsky Research Foundation

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1 Aspect-Based Sentiment Analysis: Introduction

Document-level sentiment analysis ('this review is positive') discards granular information: a restaurant review may praise the food but criticise the service. Aspect-Based Sentiment Analysis (ABSA) operates at the aspect level — 'food: positive; service: negative; ambience: neutral' — providing actionable intelligence unavailable from coarse sentiment scoring. ABSA has become one of the most commercially valuable NLP tasks precisely because of this granularity: a product manager who knows that 400 reviews mention the battery negatively but 600 mention the camera positively has far more actionable information than one who knows the product's aggregate rating is 3.8 stars.

ABSA sits at the intersection of information extraction and sentiment analysis. It requires the model to simultaneously identify what is being talked about (the aspect) and how it is being evaluated (the sentiment) — two tasks that interact and depend on each other in complex ways. Understanding ABSA requires first establishing a precise vocabulary for what 'aspect' means, which turns out to be non-trivial, and then understanding the hierarchy of sub-tasks that together define the full ABSA problem.

1.1 Aspect Categories vs. Aspect Terms

Aspect terms are explicit mentions in text ('battery life', 'screen', 'camera'). Aspect categories are predefined ontology slots ('BATTERY', 'DISPLAY', 'CAMERA#QUALITY'). The SemEval ABSA tasks distinguish these: term-level requires extraction; category-level requires inference when no explicit aspect is mentioned ('It drains fast' — BATTERY#PERFORMANCE: negative).

This distinction has significant practical consequences. A model that only detects aspect terms will fail on implicit expressions: 'the wait was over an hour' expresses negative sentiment about the SERVICE#GENERAL category, but the word 'service' never appears. Category-level ABSA requires the model to infer, from the described event or situation, which aspect category is being evaluated and what the polarity is — a task that draws on world knowledge (knowing that long waits are associated with service, not food quality) that is not present in the lexical surface of the sentence.

In practice, ABSA ontologies are domain-specific. The SemEval restaurant domain uses categories like FOOD#QUALITY, FOOD#STYLE_OPTIONS, SERVICE#GENERAL, AMBIENCE#GENERAL, and RESTAURANT#PRICES. The laptop domain uses BATTERY#OPERATION_PERFORMANCE, DISPLAY#QUALITY, LAPTOP#GENERAL, and dozens of others. The granularity and coverage of the ontology directly determines what insights an ABSA system can extract: a coarse ontology with five categories provides less actionable intelligence than a fine-grained one with fifty, but requires less labelled data to train reliably.

1.2 ABSA Sub-Tasks: ATE, ASC, ASTE, and ASQP

The ABSA problem decomposes into a hierarchy of sub-tasks, each building on the previous and together constituting the complete structured opinion extraction problem. Understanding this hierarchy is essential for selecting the right model and evaluation benchmark for a given application:

- **Aspect Term Extraction (ATE):** Given a sentence, identify all aspect term spans — the specific words or phrases that refer to an aspect being evaluated. For example, from 'The battery life is disappointing but the camera is excellent', ATE should extract ['battery life', 'camera']. ATE is a sequence labelling task (typically BIO tagging) similar to Named Entity Recognition.
- **Aspect Sentiment Classification (ASC):** Given a sentence and a known aspect term, predict the sentiment polarity toward that specific aspect. ASC assumes the aspect is already identified (either manually annotated or extracted by ATE) and focuses on correct sentiment attribution. The

challenge is that the same sentence may contain multiple aspects with different polarities, requiring the model to focus its sentiment judgment on the given aspect rather than the sentence as a whole.

- **Aspect-Sentiment Triplet Extraction (ASTE):** Extract triplets of the form (aspect term, opinion term, sentiment polarity) from a sentence in one step. The opinion term is the explicit word or phrase that expresses the opinion about the aspect ('disappointing', 'excellent' in the example above). ASTE is a more comprehensive task than ATE + ASC because it also identifies and anchors the opinion expression, not just the aspect and its polarity.
- **Aspect Sentiment Quad Prediction (ASQP):** The most comprehensive ABSA formulation, extracting quads of the form (aspect category, aspect term, opinion term, sentiment polarity). ASQP captures both the explicit aspect term (or NULL if implicit) and the aspect category from a predefined ontology, alongside the opinion term and polarity — providing fully structured, ontology-aligned output suitable for downstream analytics pipelines.

Sub-Task	Input	Output	Example
ATE	Sentence	Aspect term spans	['battery life', 'camera']
ASC	Sentence + aspect term	Polarity (pos/neg/neu)	battery life → negative
ASTE	Sentence	(aspect, opinion, polarity) triplets	('battery life', 'disappointing', neg)
ASQP	Sentence	(category, term, opinion, polarity) quads	(BATTERY#PERF, 'battery life', 'disappointing', neg)

1.3 Why ABSA Is Hard: Challenges Beyond Standard Sentiment

ABSA inherits all the challenges of sentence-level sentiment analysis (negation, sarcasm, implicit sentiment, domain shift — covered in detail in the Day 22 compendium, Section 6) and adds several additional challenges specific to the aspect-level formulation:

- **Aspect-opinion distance:** The opinion expression and the aspect term it modifies may be far apart in the sentence: 'The restaurant, despite the slow kitchen and the perpetually distracted staff, has genuinely excellent food.' Correctly attributing 'excellent' to 'food' and 'slow' to 'kitchen' requires resolving long-range syntactic dependencies that short-range n-gram features cannot capture.
- **Overlapping and nested aspects:** Some sentences contain aspects that overlap in text span or share opinion words: 'The chicken and the salmon were both overcooked.' Modelling this correctly requires tracking two separate aspect terms with the same opinion word and the same polarity — a pairing problem rather than a simple labelling problem.
- **Implicit aspects with explicit opinions:** As noted in Section 1.1, 'It drains fast' contains no explicit mention of 'battery', yet clearly refers to BATTERY#PERFORMANCE. Models must infer the aspect from the described event, requiring either a coverage-complete domain ontology or world-knowledge reasoning.
- **Neutral and mixed sentiment:** Genuine neutrality is rare and hard to distinguish from weak polarity. Mixed sentiment — 'the pasta is average but the pizza is outstanding' — requires the model to refrain from averaging across aspects and instead produce distinct labels for each one. This is precisely the difficulty that the MAMS dataset (Section 2.2) was designed to probe.

- **Cross-sentence aspects:** In longer reviews, a single aspect may be evaluated across multiple sentences: 'I ordered the sea bass. The fish itself was fresh and well-seasoned. Unfortunately it arrived completely cold.' The final negative sentiment about temperature belongs to the same FOOD#QUALITY aspect established two sentences earlier.

1.4 Relation to Day 22: From Polarity to Aspect

Day 22 covered the full spectrum of sentiment analysis methods from lexicon-based VADER and SentiWordNet through classical ML to fine-tuned BERT. Every one of those methods can serve as the ASC component in an ABSA pipeline: given a sentence and a known aspect, a sentiment classifier assigns polarity. What ABSA adds is the upstream task of identifying what to classify — the aspect extraction and pairing problem that makes ABSA substantially harder and more powerful than document-level sentiment alone. The compendium progression from Day 22 (what is the sentiment of this text?) to Day 23 (what is the sentiment about each specific thing mentioned in this text?) mirrors the progression from coarse to fine-grained opinion mining that has characterised the field's development over the past decade.

2 Datasets for ABSA

The quality, coverage, and annotation conventions of ABSA benchmarks directly determine what models can be compared and what claims can be made about progress in the field. Unlike document-level sentiment, where large datasets can be constructed by scraping star-rated reviews and treating the rating as a proxy label, ABSA annotation requires skilled human annotators to identify aspect terms, classify them into ontology categories, and label their polarity — a substantially more expensive and slower process that has constrained the size of ABSA datasets compared to those available for simpler NLP tasks.

2.1 SemEval ABSA Shared Tasks

The SemEval (Semantic Evaluation) workshop series has organised the field's most widely used ABSA benchmarks through its annual shared tasks, and understanding these datasets is prerequisite to interpreting any published ABSA result. The most significant are:

- **SemEval 2014 Task 4:** Introduced aspect-level opinion mining with two domains — restaurant reviews and laptop reviews — with annotations at both the aspect-term level and the aspect-category level. Restaurant reviews are annotated with 5 category labels (FOOD, SERVICE, AMBIENCE, PRICE, ANECDOTES/MISCELLANEOUS), each with polarity. Laptop reviews are annotated with 81 fine-grained attribute-entity categories. This dataset remains the most widely cited ABSA benchmark and serves as the standard for SemEval 2014 model comparisons appearing throughout this compendium.
- **SemEval 2015 Task 12:** Extended the 2014 task to include opinion target expression (OTE) annotations — identifying the exact text span that expresses the opinion about an aspect, not just classifying the aspect's polarity. This provides the data needed for ASTE-style models that must output opinion spans alongside aspect spans and polarity labels.
- **SemEval 2016 Task 5:** Expanded ABSA to eight languages and multiple new domains (hotels, telecom, electronics), establishing multilingual ABSA as a distinct research direction. The multilingual extension revealed that ABSA ontologies are not always directly transferable across languages — the aspect categories relevant to restaurant reviews in Arabic differ from those in English in culturally significant ways.

2.2 MAMS and Other Modern Datasets

The Multi-Aspect Multi-Sentiment (MAMS) dataset (Jiang et al., 2019) was constructed to address a significant weakness of the SemEval datasets: many SemEval sentences contain only one aspect, or aspects with the same polarity, which allows models to achieve high accuracy on ASC by simply predicting the dominant sentiment of the whole sentence rather than learning genuine aspect-level attribution. MAMS was constructed by filtering for sentences where at least two aspects have different sentiments, creating a dataset where aspect-level attention is genuinely required — models that treat the whole sentence as a positive or negative unit will fail systematically.

ARTS (Aspect Robustness Test Set, Xing et al., 2020) is an adversarially constructed complement to the SemEval 2014 restaurant dataset, designed to test whether models are genuinely performing aspect-specific sentiment attribution or are exploiting superficial patterns in the original test set. ARTS constructs challenging test cases by modifying sentence structure while preserving aspect-polarity labels, revealing that many BERT-based ABSA models that perform well on the original SemEval test set are substantially less robust on the adversarial ARTS examples — an important reminder that benchmark accuracy does not always reflect the underlying capability the benchmark was designed to measure.

2.3 Annotation Conventions and Inter-Annotator Agreement

ABSA annotation is inherently subjective in ways that exceed even standard sentiment annotation. When is 'fast service' a food service aspect vs. a restaurant attribute? Does 'the place is cosy' annotate the AMBIENCE category or the RESTAURANT#GENERAL category? Annotation guidelines for the SemEval datasets run to tens of pages precisely because so many boundary cases require explicit resolution.

Inter-annotator agreement (IAA) for ABSA tasks is typically lower than for binary polarity classification: Cohen's kappa for aspect term boundary detection typically falls in the 0.7–0.8 range; for aspect-category assignment (choosing among 80+ fine-grained categories) it can fall to 0.6 or below. This relatively low IAA is not a failure of the annotation process — it reflects genuine semantic ambiguity in the task itself. It does, however, set a natural performance ceiling: models that achieve F1 scores exceeding the human inter-annotator agreement level are likely overfitting to the specific idiosyncrasies of the test set annotation rather than demonstrating superhuman ABSA capability.

Dataset	Year	Domain	Size	Key Feature
SemEval 2014 Res.	2014	Restaurant	3,608 / 1,120 (train/test)	Aspect term + category + polarity
SemEval 2014 Lap.	2014	Laptop	3,045 / 800 (train/test)	81 fine-grained categories
SemEval 2015 Res.	2015	Restaurant	1,315 / 685 (train/test)	Opinion target expressions (OTE)
SemEval 2016 Res.	2016	Restaurant + 7 more	Multi-domain, multi-language	Multilingual ABSA
MAMS	2019	Restaurant	14,377 sentences	All sentences have conflicting aspect sentiments
ARTS	2020	Restaurant	Test set augmentation	Adversarial robustness testing

2.4 Collecting Custom ABSA Training Data

For organisations building production ABSA systems on proprietary data (enterprise customer feedback, healthcare patient comments, specialised technical reviews), the published SemEval datasets may not provide sufficient domain coverage. Constructing a custom ABSA training set is substantially more expensive than collecting document-level sentiment labels, but several strategies reduce the cost:

- **Ontology-first design:** Define the aspect ontology before annotation begins. The number of labelled examples needed scales with the number of distinct categories in the ontology — a coarser ontology (5–10 categories) requires far fewer examples to train reliably than a fine-grained one (50–80 categories). Start with the minimum granularity that supports the application's decision-making requirements.
- **Pre-annotation with existing tools:** Use an existing ABSA model (even an imperfect one) to pre-annotate the data, then have human annotators correct and verify rather than annotating from scratch. Correction is substantially faster than annotation for experienced annotators, reducing labelling time by 40–60% on well-structured review data.
- **Focused sampling:** Rather than randomly sampling from all available reviews, deliberately over-sample reviews that mention specific aspects the ontology includes, using keyword or regex filtering. This produces a more aspect-balanced training set that trains the model more efficiently

than a uniformly sampled corpus dominated by reviews that mention only one or two common aspects.

- **Active learning:** Iteratively train a model on a small initial labelled set, use it to identify the most informative unlabelled examples (those the model is most uncertain about), label only those examples, and repeat. Active learning strategies reduce the total labelling cost by 30–50% on ABSA tasks compared to random sampling while maintaining comparable model performance.

3 Pipeline vs. End-to-End ABSA Models

A foundational architectural choice in ABSA system design is whether to approach the problem as a pipeline of sequential specialised models or as a single unified model that produces structured output end-to-end. This choice affects accuracy, error propagation, training complexity, and inference latency, and understanding the trade-offs is essential for designing a production ABSA system.

3.1 Pipeline Approaches

Pipeline ABSA systems decompose the task into a sequence of individually solvable sub-problems, each handled by a separate model. A typical two-stage pipeline for ASTE operates as follows: first, an ATE model (typically a sequence labelling model, either a BiLSTM-CRF or BERT with a token classification head) identifies aspect term spans in the sentence; second, for each identified aspect span, an ASC model predicts the sentiment polarity toward that specific aspect. A three-stage pipeline additionally includes an opinion term extraction step between ATE and ASC.

The principal advantage of pipeline systems is modularity: each component can be individually trained, evaluated, and improved using its own dedicated dataset and evaluation metric. If the ASC component is underperforming, it can be retrained on additional labelled data for ASC specifically, without disturbing the ATE component. This modularity also makes it straightforward to incorporate existing components — a high-quality general-purpose NER model, for instance, can be adapted to serve as the ATE stage with domain-specific fine-tuning.

The fundamental limitation of pipeline systems is error propagation: errors made by the first component are passed on as inputs to subsequent components, where they cannot be corrected. If the ATE stage misses an aspect term or extracts an incorrect boundary, the ASC stage receives wrong input and will produce wrong output regardless of its own quality. Error propagation is especially damaging in longer pipelines and in domains where ATE is inherently noisy (implicit aspects, compound noun aspects). Empirically, end-to-end models that jointly optimise for all sub-tasks consistently outperform well-tuned pipelines on combined metrics like ASTE F1, precisely because joint training allows the model to share information between the aspect extraction and sentiment classification tasks rather than treating them as sequentially independent.

3.2 Joint and End-to-End Approaches

Joint models perform aspect extraction and sentiment classification simultaneously, sharing a single encoder and training on a combined loss function. The simplest joint formulation treats ASC as an extended sequence labelling task: each token is labelled with one of {B-POS, I-POS, B-NEG, I-NEG, B-NEU, I-NEU, O}, encoding both the aspect span boundary (B/I/O) and the sentiment polarity (POS/NEG/NEU) in a single tag set. This 'collapsed' tagging scheme is information-efficient and eliminates the pipeline error-propagation problem at the cost of a larger label space and the inability to model the interaction between sentiment context and aspect boundary detection explicitly.

More sophisticated joint models use a shared BERT encoder with separate but jointly trained heads for span detection and polarity classification, or model the task as a table-filling problem where an N×N cell grid represents all possible (head, tail) token pairs as potential aspect or opinion span relations. The table-filling or biaffine attention approaches have become particularly popular for ASTE and ASQP because they can naturally represent the pairing between aspect spans and opinion spans as a relationship between two token positions in the sentence, without requiring the pipeline to first enumerate candidate spans and then classify their relations.

3.3 Generative vs. Discriminative ABSA

A third architectural dimension, orthogonal to pipeline vs. joint, is the choice between discriminative models (which classify or label existing text) and generative models (which produce structured output text as a sequence of tokens). Generative ABSA, covered in depth in Section 4.2, uses a sequence-to-sequence model (T5, BART) to directly generate the structured output — e.g., '(battery life, disappointing, negative)' — as a text string. This approach bypasses the need to design span detection, table-filling, or classification heads entirely: the model learns the output format implicitly from training examples.

Approach	Error Propagation	Flexibility	Training Complexity	ASTE Performance
Pipeline (sequential)	High	High (modular)	Low per component	Lower (compounded errors)
Joint (multi-head)	None	Moderate	Moderate (shared encoder)	Higher
Table-filling / biaffine	None	High (captures span pairs)	Higher	State-of-the-art
Generative (T5/BART)	None	Very high (flexible output)	Moderate	Competitive, flexible

```
# Simplified joint ATE+ASC collapsed tagging scheme
# Label set: B-POS, I-POS, B-NEG, I-NEG, B-NEU, I-NEU, O

sentence  = ['The', 'battery', 'life', 'is', 'disappointing', 'but',
            'the',  'camera',  'is',  'excellent']
labels    = ['O',    'B-NEG',   'I-NEG', 'O', 'O',  'O',
            'O',    'B-POS',   'O',     'O']

# BERT token classifier with 7-class output head
# trained with cross-entropy on the collapsed label set
```

3.4 Span-Pair Extraction: Table-Filling for ASTE

The table-filling approach to ASTE, introduced by Wu et al. (2020) and refined in subsequent work, represents each sentence as an N×N grid where N is the sentence length. Each cell (i, j) in the grid represents a potential span from token i to token j. The model assigns a label to each cell indicating whether that span is an aspect term, an opinion term, or neither, and simultaneously assigns relation labels to pairs of cells indicating whether a (aspect span, opinion span) pair is a valid ASTE triplet with a specific sentiment polarity.

This formulation elegantly handles the pairing problem in ASTE: rather than first extracting all aspect spans, then extracting all opinion spans, then inferring which pairs go together (the pipeline approach), the table-filling model jointly labels span boundaries and span-pair relations in a single structured prediction task. The 2D table representation also naturally handles overlapping spans and multiple triplets within a sentence, which collapsed sequential tagging schemes handle less elegantly.

3.5 Evaluation Metrics for ABSA Models

ABSA evaluation requires metric choices that reflect the nature of the task. Standard metrics include:

- **Precision, Recall, F1 (for ATE and ASTE):** A predicted aspect span is counted as correct only if it exactly matches the annotated span (same start and end token). This exact-match criterion is strict — predicting 'battery' when the annotation is 'battery life' counts as a false positive and a false negative — and motivates models that produce complete, correctly-bounded spans rather than approximately correct ones.
- **Accuracy (for ASC):** When the aspect term is given, ASC is evaluated as multi-class accuracy over the positive/negative/neutral label set. Macro-accuracy (equal weight per class) is preferred over micro-accuracy when neutral examples are sparse and the positive/negative distinction is the primary concern.
- **F1 on complete triplets / quads (for ASTE and ASQP):** A predicted triplet (aspect, opinion, polarity) is correct only if all three elements exactly match the annotation. This strict criterion means ASTE F1 scores are typically lower than ATE F1 scores measured independently, since a correct aspect extraction followed by a wrong opinion extraction or wrong polarity counts as a failure on the ASTE metric but a success on the ATE metric.

3.6 Pre-Transformer ABSA Baselines

Before transformers became the dominant paradigm, the leading ABSA approaches combined CRF (Conditional Random Field) sequence labelling for aspect term extraction with LSTM-based sentiment classification, using pre-trained GloVe or word2vec embeddings. These pre-transformer baselines — still commonly referenced in comparison tables of published papers — provide useful context for interpreting the magnitude of the BERT-era improvement. On SemEval 2014 Restaurant ATE, the best pre-BERT systems achieved approximately 79–82% F1 (e.g., LSTM-CRF with syntactic dependency features); fine-tuned BERT raises this to 84–86% F1, a meaningful but not dramatic improvement, reflecting that ATE is primarily a sequence labelling task where careful feature engineering could compensate partially for the lack of pre-trained contextual representations.

For ASC, the pre-transformer improvement is larger: LSTM-based attention models (AT-LSTM, ATAE-LSTM, Wang et al., 2016) achieved approximately 78–82% accuracy on SemEval 2014 Restaurant ASC; fine-tuned BERT achieves 84–88%, a 6–8 point improvement that reflects BERT's substantially better contextual encoding of the interaction between the target aspect and the surrounding opinion context. This larger improvement on ASC than ATE is consistent with the general pattern that tasks requiring deep semantic interpretation of context benefit more from BERT's pre-trained representations than tasks that primarily require surface pattern recognition.

Task	LSTM-CRF / AT-LSTM	BERT (fine-tuned)	BERT-PT (domain post-trained)	Improvement
ATE (SemEval 2014 Rest.)	79–82% F1	84–86% F1	86–88% F1	+4–6 pts over LSTM
ASC (SemEval 2014 Rest.)	78–82% Acc.	84–88% Acc.	87–90% Acc.	+6–8 pts over LSTM
ASTE	~50–55% F1	~62–66% F1	~66–70% F1	+12–16 pts over LSTM

4 Transformer-Based ABSA

The introduction of BERT and its variants transformed ABSA in the same way it transformed other NLP tasks: pre-trained contextual representations replaced fixed word embeddings and recurrent encoders, producing large and consistent improvements across every ABSA sub-task and benchmark. However, standard BERT pre-trained on Wikipedia and BooksCorpus represents review text imperfectly: it has limited exposure to the vocabulary, discourse patterns, and opinion expressions characteristic of product reviews, restaurant feedback, and other opinion-rich domains. This motivated domain-adaptive pre-training strategies specifically tailored to ABSA, followed by a wave of generative approaches that reformulated ABSA as a text-generation problem rather than a classification or span-labelling task.

4.1 BERT-PT and Post-Training for ABSA

BERT-PT (Xu et al., 2019) demonstrated that post-training BERT on review-domain text before fine-tuning on ABSA tasks produces consistent improvements across SemEval benchmarks. Post-training in this context means continuing BERT's masked language model pre-training on a large corpus of unlabelled review text (Yelp restaurant reviews and Amazon laptop reviews matching the SemEval domains) before applying the standard ABSA fine-tuning. This domain-adaptive pre-training (DAPT) approach, covered conceptually in the Day 20 compendium (Section 7), adapts BERT's vocabulary representations toward the specific lexical patterns of review text — terms like 'crispy', 'overcooked', 'crunchy' for food reviews; 'sluggish', 'overheats', 'responsive' for laptop reviews — without requiring any labelled ABSA data in the post-training stage.

The improvements from domain-adaptive post-training are additive to improvements from task-specific fine-tuning: BERT-PT followed by ABSA fine-tuning outperforms both standard BERT fine-tuned on ABSA data and domain-adapted BERT applied zero-shot. The magnitude of improvement is largest for tasks and aspects requiring fine-grained domain vocabulary — ATE benefits more from domain post-training than ASC, because correctly identifying aspect term boundaries depends heavily on recognising domain-specific noun phrases that standard BERT's vocabulary representations may conflate or undersegment.

Aspect-Aware Attention

Beyond domain-adaptive pre-training, several BERT-based ABSA architectures augment the standard BERT encoder with explicit aspect-aware attention mechanisms. The core insight is that standard BERT self-attention is not specifically directed toward the target aspect when performing ASC: the model may attend broadly to the whole sentence rather than focusing on the words most relevant to the given aspect's sentiment. Aspect-aware attention injects the aspect term representation as an additional query or context signal into selected BERT layers, guiding attention toward the region of the sentence most relevant to that specific aspect.

```
# BERT-based ASC: aspect-conditioned sentence encoding
# Input format: [CLS] sentence [SEP] aspect_term [SEP]
# The model attends to aspect_term context while encoding the sentence

from transformers import AutoTokenizer, AutoModelForSequenceClassification

tokenizer = AutoTokenizer.from_pretrained('bert-base-uncased')
model = AutoModelForSequenceClassification.from_pretrained(
    'bert-base-uncased', num_labels=3)  # positive / negative / neutral
```

```
sentence = 'The battery life is disappointing but the camera is excellent.'
aspect   = 'battery life'

# Pack sentence + aspect as a sentence pair (BERT segment A + segment B)
inputs = tokenizer(sentence, aspect, return_tensors='pt',
                   truncation=True, max_length=128)
logits = model(**inputs).logits  # classify [CLS] rep for this aspect
```

4.2 Generative ABSA with T5

The generative approach to ABSA, introduced in GAS (Generative ABSA Solver, Zhang et al., 2021) and subsequently developed in Paraphrase-based ABSA, Seq2Path, and related work, reframes ABSA not as a classification or span-labelling task but as a conditional text generation task: given a sentence, generate a structured text string that encodes the complete ABSA output. For ASTE, the model is trained to produce strings of the form 'aspect: battery life; opinion: disappointing; sentiment: negative | aspect: camera; opinion: excellent; sentiment: positive' from the input sentence.

The generative paradigm has several practical advantages over discriminative approaches. It handles all ABSA sub-tasks within a single unified model and training procedure, with no need for separate heads for span detection, pair classification, or polarity labelling. It naturally handles multi-aspect sentences by generating a variable number of structured tuples, whereas discriminative models must pre-specify the maximum number of aspect-sentiment pairs or enumerate all candidate spans. The output format can be adapted to different ABSA formulations (ATE, ASC, ASTE, ASQP) simply by changing the generation target string, without modifying the model architecture.

Research Spotlight: Generative ABSA with T5 (Zhang et al., 2021)

Rather than training separate models for aspect extraction and sentiment classification, GAS (Generative ABSA Solver) prompts T5 to directly generate structured output: 'aspect: food, sentiment: positive; aspect: service, sentiment: negative'. This unified generative approach surpasses pipeline models on all SemEval ABSA sub-tasks.

```
# Generative ABSA with T5 (GAS formulation)
from transformers import T5ForConditionalGeneration, T5Tokenizer

model = T5ForConditionalGeneration.from_pretrained('t5-base')
tokenizer = T5Tokenizer.from_pretrained('t5-base')

# Input: task prefix + sentence
input_text = 'absa triplet: The battery life is disappointing but the camera is excellent.'
input_ids = tokenizer(input_text, return_tensors='pt').input_ids

# Target: structured output string (during training)
target_text = '(battery life, disappointing, negative) | (camera, excellent, positive)'
labels = tokenizer(target_text, return_tensors='pt').input_ids
```

```
# At inference, model.generate() produces the structured output directly
output_ids = model.generate(input_ids, max_length=128)
print(tokenizer.decode(output_ids[0], skip_special_tokens=True))
```

The primary limitation of generative ABSA is output validity: a generative model can produce malformed output strings that do not parse into valid aspect-sentiment tuples, especially for complex sentences with many aspects or unusual aspect expressions. Post-processing validity checks and constrained decoding (restricting generation to only produce tokens that result in valid structured output) are standard mitigations in production generative ABSA systems. Despite this limitation, generative ABSA models have become the dominant approach in recent research and achieve state-of-the-art on the ASQP task in particular, where the combination of aspect categories, aspect terms, opinion terms, and polarity labels makes discriminative multi-head architectures cumbersome.

4.3 Comparing Transformer ABSA Approaches

Each transformer-based ABSA architecture makes distinct trade-offs between modelling power, training data requirements, and inference practicality. The table below summarises the key choices for the four approaches most commonly used in production and research settings:

Approach	**Task Coverage**	**Training Data Required**	**Output Validity**	**Best SemEval ATE F1**
BERT token classifier (joint)	ATE + ASC	Thousands of ABSA-labelled sentences	Always valid (label set is fixed)	~84–86%
BERT-PT (domain post-trained)	ATE + ASC + (ASTE with span head)	Domain corpus + ABSA labels	Always valid	~86–88%
Table-filling / biaffine	ASTE, ASQP	Thousands of triplet-labelled sentences	Always valid	~75–78% (ASTE F1)
Generative T5 (GAS)	ATE, ASC, ASTE, ASQP (unified)	Thousands of labelled sentences	May produce malformed output	~73–76% (ASTE F1)

Practical deployment considerations often favour BERT-PT with a joint token classifier over purely generative approaches for high-throughput production ABSA (customer feedback processing at scale), because the discriminative model's output is always structurally valid and its inference latency is lower. Generative models become the preferred choice when the application requires the most comprehensive sub-task coverage (ASQP with all four output dimensions) or when the output format must be adaptable without retraining the model architecture.

4.4 End-to-End Training Pipeline for Transformer ABSA

The following walkthrough outlines the complete training pipeline for a BERT-based joint ATE+ASC model on SemEval 2014 data, connecting the architectural discussions above to concrete implementation steps relevant to the Gudsky team's own ABSA projects:

```
from transformers import AutoTokenizer, AutoModelForTokenClassification
from transformers import TrainingArguments, Trainer
from datasets import load_dataset
import numpy as np

# Label set: B-POS, I-POS, B-NEG, I-NEG, B-NEU, I-NEU, O  (7 classes)
LABELS = ['O','B-POS','I-POS','B-NEG','I-NEG','B-NEU','I-NEU']
label2id = {l: i for i, l in enumerate(LABELS)}

tokenizer = AutoTokenizer.from_pretrained('bert-base-uncased')
model = AutoModelForTokenClassification.from_pretrained(
    'bert-base-uncased', num_labels=len(LABELS), id2label={i:l for i,l in enumerate(LABELS)}
)

# Tokenise with word-to-token alignment for BIO label propagation
def tokenise_and_align_labels(batch):
    enc = tokenizer(batch['tokens'], is_split_into_words=True,
                    truncation=True, max_length=128)
    aligned = []
    for i, labels in enumerate(batch['ner_tags']):
        word_ids = enc.word_ids(batch_index=i)
        aligned.append([label2id.get(labels[wid], -100)
                        if wid is not None else -100 for wid in word_ids])
    enc['labels'] = aligned
    return enc

training_args = TrainingArguments(
    output_dir='./absa_joint', num_train_epochs=10,
    per_device_train_batch_size=16, learning_rate=2e-5,
    weight_decay=0.01, warmup_ratio=0.1,
)
```

ABSA on Yelp Reviews

A restaurant with 4.2/5 stars might have: food=4.8, service=3.1, ambience=4.5, price=3.8.

ABSA on 50,000 Yelp reviews reveals this granularity. Tripadvisor's product recommendations now surface aspect-level ratings from review mining rather than single star scores — dramatically improving booking conversion rates.

5 Emotion Recognition in Text

Emotion recognition in text — identifying the discrete emotional state expressed by or elicited in a piece of text — extends the binary or ternary polarity classification of sentiment analysis to a richer categorical space. Where sentiment analysis asks 'is this positive, negative, or neutral?', emotion recognition asks 'is this joyful, angry, fearful, sad, surprised, or disgusted?' — or in more fine-grained formulations, 'does this express anticipation, trust, annoyance, apprehension, admiration, or remorse?'. This richer representation captures distinctions that matter for downstream applications: a customer who is angry requires a different response than one who is sad, even though both are expressing negative sentiment.

Emotion recognition is related to but distinct from affective computing (the broader field of detecting and responding to human emotional states across modalities including physiological signals, facial expressions, and speech prosody) and from opinion mining (which focuses on the evaluative dimension of sentiment). Text-based emotion recognition specifically operates on written language, drawing on linguistic cues including explicit emotion words ('I am furious'), physiological metaphors ('my heart sank'), and event descriptions whose emotional valence is inferred from world knowledge.

5.1 Ekman's 6 Basic Emotions

Paul Ekman's theory of basic emotions, developed from cross-cultural studies of facial expression recognition in the 1960s–1980s, proposed that six emotions are universally expressed and recognised across cultures: happiness, sadness, anger, fear, disgust, and surprise. This taxonomy became the dominant categorical framework for text-based emotion recognition research in the 2000s and 2010s, and most early emotion recognition datasets, lexicons, and models used some variant of these six categories.

Ekman's six-emotion taxonomy has both practical and theoretical justifications for computational use: six categories are few enough to be reliably labelled by annotators and to permit accurate classification by supervised models trained on datasets of feasible size; the categories are semantically distinct enough that inter-annotator agreement is typically higher than for more fine-grained emotion taxonomies; and the universal-expression claim provides a theoretical grounding for applying models trained on English data to other languages (though this cross-lingual transfer assumption is contested in more recent cross-cultural emotion research). The SemEval 2018 Task 1 (Affect in Tweets) used a subset of Ekman's categories as labels; the ISEAR (International Survey on Emotion Antecedents and Reactions) dataset is labelled with seven categories very close to Ekman's six; and many biomedical and clinical emotion recognition datasets use Ekman-derived label sets.

The principal limitation of a six-category framework is insufficient resolution for many applications. Customer service applications need to distinguish frustrated from confused; mental health monitoring needs to distinguish anxious from depressed; creative writing assistance needs to distinguish melancholic from devastated. This limitation motivated the development of richer taxonomies, including Plutchik's wheel (8 primary emotions with degrees of intensity and composite emotions), and fine-grained datasets like GoEmotions (27 categories) discussed in the Day 22 compendium (Section 10.3).

Emotion	Ekman Category	Example Text	Key Distinguishing Linguistic Cues
Happiness	Basic	I am absolutely thrilled about this!	Joy words, exclamation, intensifiers
Sadness	Basic	I feel so lost and hopeless.	Loss, futility words, downward metaphors

Anger	Basic	This is completely unacceptable!	Blame, injustice, intensifiers, exclamation
Fear	Basic	I don't know what's going to happen.	Uncertainty, threat, future orientation
Disgust	Basic	That was absolutely revolting.	Revulsion words, violation metaphors
Surprise	Basic	I had no idea this was happening!	Unexpectedness, exclamation, contrast

5.2 NRC Emotion Lexicon

The NRC Emotion Lexicon (EmoLex, Mohammad & Turney, 2013) is the most widely used lexical resource for text-based emotion recognition, constructed through crowdsourced annotation of over 14,000 English words with association scores for Ekman's six basic emotions plus two additional dimensions: positive and negative polarity. Each word is binary-labelled for each of the eight categories (present/absent), based on responses from Mechanical Turk workers who answered the question 'does this word evoke [emotion]?'. The resulting resource associates words with their typical emotional connotation rather than their literal semantic category: 'prison' evokes fear and anger even though it is not itself an emotion word.

The NRC lexicon has been translated into over 100 languages via word-level alignment through translation services, producing the NRC Word-Emotion Association Lexicon in multiple languages (NRC EmoLex multilingual). While machine-translation-based lexicon construction inevitably introduces noise (translating emotional connotations across languages and cultures is not straightforward), the multilingual NRC lexicon provides a practical starting point for emotion recognition in languages without independently constructed lexical resources.

```
# Using NRC Emotion Lexicon for text emotion scoring
import pandas as pd
from collections import defaultdict

# Load NRC lexicon (word, emotion, association: 0 or 1)
nrc = pd.read_csv('NRC-Emotion-Lexicon-Wordlevel-v0.92.txt',
                  sep='\t', header=None, names=['word','emotion','assoc'])
nrc_dict = nrc[nrc['assoc']==1].groupby('word')['emotion'].apply(list).to_dict()

def score_emotions(text):
    counts = defaultdict(int)
    for word in text.lower().split():
        for emotion in nrc_dict.get(word, []):
            counts[emotion] += 1
    return dict(counts)

text = 'I am absolutely furious and deeply saddened by what happened.'
print(score_emotions(text))
# {'anger': 2, 'negative': 3, 'sadness': 1, 'fear': 1, ...}
```

The NRC Valence, Arousal, and Dominance (VAD) lexicon complements the categorical EmoLex with continuous dimensional scores for each word across the three affective dimensions discussed in Section

5.4. Having both categorical and dimensional representations available from the same source enables hybrid models that use categorical labels for classification tasks and VAD scores for intensity regression — a combination particularly useful in customer service and mental health applications where both the type and the intensity of the detected emotion drive downstream decisions.

Lexicon	Coverage	Representation	Key Application
SentiWordNet (Day 22)	~117K WordNet synsets	Pos/Neg/Obj scores	General formal-text sentiment
VADER (Day 22)	~7,500 tokens + rules	Compound score [−1,+1]	Social media sentiment
NRC EmoLex	~14,000 words	8 binary emotion categories	Categorical emotion detection
NRC VAD Lexicon	~20,000 words	Valence, Arousal, Dominance scores	Dimensional emotion intensity
SenticNet	~100K concepts	Polarity + primary emotion	Commonsense-grounded ABSA

5.3 Deep Learning for Emotion Recognition

Deep learning approaches to emotion recognition follow the same architectural progression as sentiment analysis (Day 22, Sections 3–5): from CNN and BiLSTM models using pre-trained word embeddings, through BERT fine-tuned for multi-label emotion classification, to large language models used with chain-of-thought prompting for complex or ambiguous emotion reasoning. Emotion recognition adds several dimensions of complexity beyond binary sentiment classification that influence architectural choices:

- **Multi-label classification:** Unlike polarity classification (one label per sentence), emotion recognition is typically multi-label: a text can simultaneously express both joy and surprise, or both anger and disgust. This requires either sigmoid-based multi-label outputs rather than softmax, or a threshold-based post-processing step on softmax outputs.
- **Fine-grained vs. coarse-grained:** Classifying 6 basic emotions requires less model capacity and fewer training examples than classifying 27 fine-grained emotions. Fine-grained classification benefits substantially from hierarchical label structures (e.g., 'annoyance' and 'rage' both belong to the 'anger' family) that can be incorporated via hierarchical loss functions or label-hierarchy-aware attention.
- **Intensity regression:** Some applications require predicting not just which emotion is present but its intensity (slightly angry vs. enraged). SemEval 2018 Task 1 formalised this as a regression task alongside classification, with continuous intensity scores on a 0–1 scale. BERT-based models trained with MSE regression heads outperform classification-then-intensity-labelling approaches on intensity prediction.

The GoEmotions dataset (Demszky et al., 2020), with 58,000 Reddit comments labelled across 27 emotion categories, established the current standard benchmark for fine-grained emotion recognition. Fine-tuned BERT achieves macro-F1 of approximately 0.46 on GoEmotions — substantially lower than the 0.7+ F1 achievable on coarser Ekman-style benchmarks, reflecting both the genuine difficulty of fine-grained emotion distinction and the lower inter-annotator agreement at finer granularity. This performance gap motivates hierarchical classification approaches that first predict the coarse emotion family before predicting the fine-grained category within that family.

5.4 Plutchik's Wheel and Dimensional Emotion Models

While Ekman's discrete category model has dominated computational work, a complementary tradition in affective science models emotion as a continuous space with two or three dimensions rather than a discrete set of categories. The circumplex model of affect (Russell, 1980) characterises every emotional state by its position on two axes: valence (negative to positive) and arousal (low to high). Anger is high-arousal negative; sadness is low-arousal negative; excitement is high-arousal positive; contentment is low-arousal positive. Plutchik's wheel of emotions (1980) organises eight primary emotions in a circle by similarity, with opposite emotions at 180 degrees and intensity varying along the radial axis.

Dimensional models have practical advantages for applications that require continuous or graded emotion representation rather than discrete labels. A sentiment intensity score on a valence-arousal plane, for instance, can be directly used to rank customer feedback by urgency (high-arousal negative = most urgent for human follow-up) in a way that a three-class positive/negative/neutral label cannot. The VAD (Valence, Arousal, Dominance) lexicon (Mohammad, 2018), containing valence, arousal, and dominance scores for over 20,000 English words, provides a lexical resource for dimensional emotion analysis analogous to SentiWordNet's role in categorical sentiment analysis.

5.5 Emotion Recognition in Dialogue

Emotion Recognition in Conversation (ERC) extends single-utterance emotion recognition to multi-turn dialogue, where the emotion of each utterance is interpreted in the context of the conversational history. The IEMOCAP (Interactive Emotional Dyadic Motion Capture) dataset and the DailyDialog dataset are the primary benchmarks for ERC. ERC models typically encode conversational context using recurrent or transformer-based context encoders that maintain a conversation history representation, with emotion predictions for each turn conditioned on both the current utterance and the history.

```
# Fine-tuning BERT for multi-class emotion classification (GoEmotions)
from transformers import AutoTokenizer, AutoModelForSequenceClassification
import torch

EMOTIONS = ['admiration','amusement','anger','annoyance','approval',
            'caring','confusion','curiosity','desire','disappointment',
            'disapproval','disgust','embarrassment','excitement','fear',
            'gratitude','grief','joy','love','nervousness',
            'optimism','pride','realisation','relief','remorse',
            'sadness','surprise']  # 27 GoEmotions categories

model = AutoModelForSequenceClassification.from_pretrained(
    'bert-base-uncased',
    num_labels=len(EMOTIONS),
    problem_type='multi_label_classification',  # multiple emotions per text
)
# Loss: BCEWithLogitsLoss; threshold 0.3 per label at inference
```

6 Multimodal Sentiment and Emotion

Human emotional communication is inherently multimodal: we express and perceive emotion through the words we choose, the tone and rhythm of our speech, our facial expressions, our body posture, and our gestures, simultaneously. Text-only sentiment and emotion analysis captures only one channel of this rich signal. Multimodal sentiment and emotion analysis integrates text, acoustic speech features, and visual features (facial action units, body pose) into a joint model that can leverage cross-modal consistency and complementarity to make more accurate and robust predictions than any single modality alone.

6.1 Datasets: CMU-MOSI and CMU-MOSEI

The CMU-MOSI (Multimodal Opinion Sentiment and Intensity, Zadeh et al., 2016) and CMU-MOSEI (Multimodal Opinion Sentiment and Emotion Intensity, Zadeh et al., 2018) datasets are the field's primary benchmarks for multimodal sentiment and emotion analysis. CMU-MOSI contains 2,199 opinionated video segments from YouTube, each annotated with a continuous sentiment intensity score on a [−3, +3] scale. CMU-MOSEI extends this to 23,453 segments with both sentiment scores and per-segment emotion labels (Ekman's six emotions). Both datasets provide time-aligned text transcripts, acoustic features extracted from audio (MFCCs, pitch, energy), and visual features extracted from video (facial action units via OpenFace, pose estimation).

A key empirical finding from work on these datasets is that text is the dominant modality for sentiment prediction in most segments — acoustic and visual features individually contribute less predictive power than text alone — but their combination with text produces consistent improvements of 2–5 percentage points in accuracy over text-only models. This finding suggests that acoustic and visual signals primarily serve as confirmatory or disambiguating signals rather than as independent carriers of sentiment information: they provide evidence that helps the model resolve ambiguous text cases rather than providing distinct sentiment channels.

6.2 Fusion Architectures

Multimodal fusion refers to the mechanism by which information from different modalities is combined in the model. Three broad fusion strategies are standard in the literature:

- **Early fusion (feature-level):** Concatenate the feature representations from all modalities at the input level, before any deep processing, and train a single model on the concatenated multi-modal feature vector. Simple to implement but assumes that the modalities are best processed jointly from the start, and can be sensitive to feature scale differences across modalities.
- **Late fusion (decision-level):** Train separate unimodal models for each modality, then combine their output predictions (probabilities or logits) via averaging, voting, or a learned fusion layer. Easy to modularise and allows each unimodal model to be optimised independently, but does not capture cross-modal interactions that require seeing multiple modalities simultaneously.
- **Cross-modal attention (intermediate fusion):** The dominant approach in recent work. Transformer-based cross-modal attention mechanisms allow text representations to attend to acoustic and visual representations and vice versa, learning contextual alignments between modalities that neither early nor late fusion can capture. Models like MulT (Multimodal Transformers, Tsai et al., 2019) and later work using unified multimodal Transformer encoders achieve state-of-the-art on CMU-MOSI and CMU-MOSEI through cross-modal attention.

```
# Conceptual cross-modal attention for text + audio fusion
```

```python
# Text tokens attend to audio frames and vice versa
import torch.nn as nn

class CrossModalAttention(nn.Module):
    def __init__(self, d_text=768, d_audio=74, d_model=256):
        super().__init__()
        # Project each modality to shared dimension
        self.text_proj  = nn.Linear(d_text, d_model)
        self.audio_proj = nn.Linear(d_audio, d_model)
        self.attn = nn.MultiheadAttention(d_model, num_heads=8, batch_first=True)

    def forward(self, text_feats, audio_feats):
        q = self.text_proj(text_feats)   # text queries
        k = self.audio_proj(audio_feats) # audio keys
        v = self.audio_proj(audio_feats) # audio values
        attended, _ = self.attn(q, k, v) # text attends to audio
        return attended
```

6.3 Multimodal Emotion in Speech

Speech emotion recognition (SER) — classifying the emotion in a spoken utterance using acoustic features — predates multimodal approaches and represents a rich parallel field to text-based emotion recognition. Acoustic features most predictive of emotion include fundamental frequency (F0/pitch) contour, energy and intensity patterns, speaking rate, voice quality measures (jitter, shimmer, harmonics-to-noise ratio), and spectral features (MFCCs, mel spectrogram). Anger typically produces elevated pitch and energy; sadness produces lower pitch, slower speaking rate, and reduced energy; fear produces variable pitch and elevated speaking rate.

Modern SER systems use end-to-end deep learning on raw audio waveforms or mel spectrograms, with CNN or Transformer-based encoders learning acoustic representations directly from data rather than using hand-engineered acoustic features. Wav2Vec 2.0 (Baevski et al., 2020) and its successors provide pre-trained speech representations analogous to BERT's pre-trained language representations, and fine-tuning Wav2Vec-based models on labelled SER datasets achieves state-of-the-art performance on standard benchmarks like IEMOCAP (Interactive Emotional Dyadic Motion Capture) and MSP-IMPROV.

Modality	Key Features	Strengths	Limitations
Text	Word semantics, syntax, sentiment lexicon	High information density; well-studied	Misses prosodic and visual cues
Audio / Speech	Pitch, energy, speaking rate, MFCCs	Captures prosodic emotion markers	Noisy in non-studio conditions
Video / Visual	Facial action units, pose, gaze	Captures non-verbal signals	Computationally expensive; privacy concerns
Multimodal (all)	Fused cross-modal representation	Most robust; handles unimodal ambiguity	Data collection complexity; alignment

6.4 Temporal Alignment in Multimodal Analysis

A challenge specific to multimodal analysis of video data is temporal alignment: text transcripts, audio signals, and video frames operate at different time scales and must be aligned before a joint model can process them. A spoken sentence may take 3–5 seconds; a facial expression may peak and fade in 0.5 seconds within that interval; the acoustic prosody of a single word may be meaningful independent of the full sentence. Forced alignment tools (such as Montreal Forced Aligner) align word-level text transcripts to audio time segments; face tracking and action unit extraction tools (OpenFace, MediaPipe) produce frame-level visual features.

Most multimodal sentiment models simplify the alignment problem by working at the utterance level — aggregating audio and visual features across the duration of a single utterance and pairing those aggregated features with the utterance's text — rather than at the sub-word or frame level. This simplification loses temporal fine structure but is substantially more robust to alignment errors and noise in the speech and visual streams. Utterance-level alignment is the dominant approach in CMU-MOSI and CMU-MOSEI experiments, and the state-of-the-art results on those benchmarks are all obtained with utterance-level models.

6.5 Privacy and Ethical Constraints on Multimodal Emotion Data

Collecting and using video and audio data for emotion analysis raises significant privacy and consent considerations that text-only sentiment analysis largely avoids. Facial expression analysis in particular involves processing biometric data (facial geometry) that is explicitly regulated under GDPR's special category data provisions in the European Union and equivalent legislation in other jurisdictions. Any multimodal emotion analysis system deployed on human subjects — including workplace monitoring, customer-facing video interactions, or therapy session analysis — requires explicit consent, data minimisation (processing only the emotion-relevant features, not storing raw video), and clear governance about who can access the derived emotional inferences and for what purposes.

6.6 Multimodal ABSA: Extending Aspect Analysis Beyond Text

An emerging research direction combines ABSA's aspect-level granularity with multimodal inputs. In product review videos on YouTube or TikTok, a reviewer may express positive sentiment verbally ('the screen is gorgeous') while their facial expression and vocal prosody convey genuine enthusiasm that amplifies the textual signal, or may describe a product feature in neutral language while acoustic cues (hesitation, falling intonation) reveal underlying dissatisfaction. Multimodal ABSA extracts aspect-level sentiment across all available modalities simultaneously, producing more accurate and reliable aspect-polarity pairs than text-only ABSA for video review content.

The multimodal ABSA benchmarks are significantly smaller than the text-only SemEval datasets, reflecting the additional cost of collecting and annotating video review content at the aspect level. The MASAD (Multimodal Aspect-level Sentiment Analysis Dataset, Zhao et al., 2021) and Twitter-15/17 multimodal datasets (containing tweet text paired with images) are the primary current benchmarks. Performance on these datasets is still substantially below text-only ABSA benchmarks due to the smaller training sets and the additional complexity of aligning cross-modal aspect expressions — making multimodal ABSA an area with significant room for improvement and active research opportunity.

Dataset	Modalities	Task	Size
CMU-MOSI	Text + Audio + Video	Document sentiment (continuous)	2,199 segments
CMU-MOSEI	Text + Audio + Video	Sentiment + 6-class emotion	23,453 segments

IEMOCAP	Text + Audio + Video	4-class / 6-class emotion in dialogue	7,380 utterances
MASAD	Text + Image	Aspect-level sentiment	~8,000 pairs
Twitter-15/17	Text + Image	Aspect-level sentiment (tweets)	~4,500 / 4,000 pairs

7 Applications: Customer Feedback, Mental Health

The ABSA and emotion recognition techniques covered in Sections 1–6 find their highest-impact applications in two broad domains: customer feedback analysis (where ABSA provides granular product and service insights that drive business decisions) and mental health NLP (where emotion recognition provides clinical decision support and early warning signals). These two application domains differ fundamentally in their stakes and ethical requirements — misclassifying a restaurant aspect as positive is a minor quality issue; misclassifying distress signals in a therapy transcript could delay critical care — but both benefit from the same underlying advances in fine-grained opinion and emotion extraction.

7.1 Customer Feedback Analysis at Scale

Customer feedback arrives through multiple channels simultaneously: product reviews, social media posts, customer service chat transcripts, post-purchase surveys, Net Promoter Score (NPS) open-text responses, and app store reviews. A production customer feedback analytics system must ingest, process, and extract structured insights from all of these channels, each with its own text distribution, vocabulary, and review structure. ABSA provides the crucial capability of decomposing aggregate sentiment into aspect-level insights that are directly actionable: rather than 'NPS comments are trending negative this quarter', a business unit receives 'comments mentioning CHECKOUT_EXPERIENCE have dropped 15% in positive sentiment since the UI redesign, while PRODUCT_QUALITY sentiment is unchanged'.

The implementation of a production ABSA feedback system involves several components beyond the core model: a domain-specific ontology definition (what aspects are relevant for this product category and what granularity is appropriate?), a data collection and labelling pipeline (how are new review sources onboarded and validated?), a model monitoring component (tracking aspect-level model performance on held-out labelled data, given the domain drift and language evolution discussed in Day 22), and a visualisation and alerting layer that presents aspect-level sentiment trends to business stakeholders in an interpretable format.

7.2 Mental Health NLP

The application of NLP to mental health monitoring and support is one of the most active and ethically significant areas at the intersection of AI and healthcare. Text produced by individuals with mental health conditions — in clinical settings (therapy transcripts, clinical notes), research settings (posts on Reddit's r/depression or r/SuicideWatch), and everyday digital life (social media, messaging apps) — contains linguistic signals that correlate with specific conditions, symptom severity, and risk levels. Emotion recognition is central to many of these signals: elevated anger and decreased positive affect in social media language are associated with depression; elevated fear and anticipatory language are associated with anxiety disorders; specific patterns of emotional expression change have been linked to manic episodes in bipolar disorder.

Use Case: Mental Health Emotion Detection

DARPA's WARCAT programme used emotion recognition NLP on veteran therapy session transcripts to identify

emotional patterns associated with PTSD episodes. The system flags sessions showing elevated fear/sadness

co-occurrence for clinical review — tested at Walter Reed National Military Medical Centre.

Beyond clinical surveillance, NLP-based mental health tools include: crisis detection systems that identify suicidal ideation in text (applied by social media platforms to direct at-risk users to crisis resources); digital phenotyping research that characterises mental health conditions from the passive observation of digital behaviour including language patterns; and mental health chatbots that provide psychoeducation and first-line emotional support, using emotion recognition to adapt their responses to the user's detected emotional state.

7.3 Ethical Considerations in Mental Health NLP

The application of emotion recognition and sentiment analysis to mental health contexts carries significant ethical weight that deserves explicit treatment beyond what is typically required for business sentiment applications. Several specific concerns are well-established in the literature and directly relevant to any research team in this space:

- **Consent and privacy:** Analysing social media posts or clinical transcripts for mental health signals requires careful attention to consent. Research using public social media data may be technically legal but ethically contested when subjects are unaware their mental health can be inferred from their posts. Clinical NLP applications require institutional review board approval and compliance with patient data protection regulations.
- **Label reliability and stigma:** Mental health labels in training datasets (depression, anxiety, PTSD) are typically derived from self-report or community membership (e.g., posting in r/depression) rather than clinical diagnosis. Models trained on these proxy labels will be calibrated to those proxies, not to diagnostic criteria — a distinction with potentially serious consequences in clinical deployment contexts.
- **Model failures in high-stakes settings:** A false positive from a brand sentiment model costs a minor investigation; a false positive from a PTSD detection model could lead to unnecessary clinical intervention; a false negative could fail to flag a genuine crisis. The asymmetric consequences of different error types in mental health applications require explicit specification of precision-recall trade-offs in deployment, and ongoing human oversight of model outputs rather than automated-only decision-making.
- **Fairness across demographic groups:** Mental health expression patterns differ across cultures, genders, and age groups. A model trained predominantly on English-language Western social media data may perform systematically differently across demographic groups, an equity concern that must be explicitly evaluated before clinical deployment.

7.4 Production ABSA System Architecture

A production customer feedback ABSA system combines several components beyond the core model. Understanding how these components fit together is important for anyone building or evaluating such a system:

```
# Simplified production ABSA feedback pipeline

# Stage 1: Ingestion & preprocessing
reviews = ingest_from_sources(['app_store', 'play_store', 'yelp', 'support_chat'])
reviews = [preprocess(r) for r in reviews]   # normalise, dedup, language detect

# Stage 2: ABSA inference
for review in reviews:
    triplets = absa_model.extract(review['text'])  # returns [(aspect, opinion, polarity)]
    review['aspects'] = map_to_ontology(triplets)  # e.g. 'battery life' -> BATTERY#PERF
```

```
# Stage 3: Aggregation & trend tracking
dashboard = aggregate_by_aspect(reviews, time_window='7d')
alerts = [a for a in dashboard if a['delta_negative'] > ALERT_THRESHOLD]

# Stage 4: Monitoring & drift detection
monitor_model_performance(absa_model, labelled_holdout_set, schedule='weekly')
```

Key operational considerations that determine whether a production ABSA system provides lasting business value include: ontology governance (the aspect ontology must be actively maintained as products and services evolve, with a clear process for adding, merging, and retiring categories); model retraining cadence (sentiment about new product features or service changes may not be well-represented in the original training data, requiring periodic re-annotation and retraining); and stakeholder calibration (business users often interpret aspect-level scores as absolute quality ratings rather than relative sentiment measures, so clear documentation of what the scores mean and how they should be compared is essential).

7.5 Selecting the Right ABSA/Emotion Approach by Application

Different application domains impose different constraints on which ABSA and emotion recognition techniques are appropriate. The following table maps the major application categories to the recommended approach, the key technical requirement that drives that choice, and the primary risk to manage:

Application	Recommended Approach	Key Requirement	Primary Risk
E-commerce review mining	BERT-PT + ASTE (triplet)	Granular product insights	Ontology staleness as products evolve
Real-time social media monitoring	DistilBERT / BERTweet (fast)	Low latency, high throughput	Language drift; sarcasm misclassification
Contact centre VoC	BERT-based ABSA + aspect trends	Actionable agent coaching signals	Privacy; transcript quality from ASR
Mental health monitoring (research)	BERT multi-label emotion + LIWC	Sensitivity to distress signals	Label proxy quality; demographic fairness
Mental health monitoring (clinical)	Validated clinical NLP + human review	Regulatory compliance; explainability	False negatives in crisis detection
Multimodal product reviews (video)	Cross-modal Transformer + ABSA head	Aspect-level across text + audio/visual	Small multimodal ABSA dataset sizes

8 Research: Commonsense Reasoning for ABSA

The frontier of ABSA research in 2024–2026 is moving beyond purely data-driven pattern matching toward models that incorporate structured commonsense knowledge to handle the implicit aspects, indirect sentiment expressions, and complex multi-clause opinions that continue to challenge even the best fine-tuned transformer models. This final section surveys three related research directions: commonsense knowledge integration for ABSA, large language model approaches to zero-shot and few-shot ABSA, and the emerging area of conversational ABSA.

8.1 Commonsense Knowledge Integration

Many ABSA challenges reduce to a commonsense reasoning problem. Correctly attributing 'It drains fast' to BATTERY#PERFORMANCE: negative requires knowing that 'drains' in the context of a device review refers to battery drain, that fast battery drain is a negative property of a device, and that BATTERY#PERFORMANCE is the relevant aspect category for this type of property. None of this knowledge is present in the text itself — it must be supplied by a knowledge source or inferred from pre-training.

SenticNet (Cambria et al., 2020) is a commonsense knowledge base specifically designed for sentiment and opinion reasoning, containing over 100,000 natural language concepts with associated affective properties: polarity, primary emotion (Ekman categories), secondary emotion (Plutchik extended categories), and semantic primitives. SenticNet differs from lexicons like SentiWordNet and VADER in that it operates at the concept level rather than the word level ('battery life' as a concept, 'fast drain' as a concept) and encodes the contextual conditions under which a concept is positive or negative.

ConceptNet (Speer et al., 2017) is a broader commonsense knowledge graph encoding general world knowledge relationships: 'battery' IsA 'power source', 'drain fast' CausesDesire 'replace battery', 'short battery life' HasProperty 'inconvenient'. ABSA models that incorporate ConceptNet embeddings or ConceptNet-enriched attention mechanisms can leverage these relationships to correctly handle implicit aspects and indirect opinion expressions that purely pattern-based models miss.

The practical integration of commonsense knowledge into transformer ABSA models has been achieved through several mechanisms: knowledge-enriched embeddings that augment BERT token representations with entity and relation embeddings from the knowledge graph; graph attention networks that propagate sentiment and semantic information across ConceptNet subgraphs rooted at the aspect and opinion terms in the input sentence; and pre-training on knowledge-intensive objectives (such as predicting ConceptNet relations for pairs of words in a sentence) before fine-tuning on ABSA tasks. Each mechanism provides a different trade-off between the depth of knowledge integration and the complexity of the training pipeline.

Knowledge Source	Type	Coverage	Best For in ABSA
SenticNet	Affective knowledge graph	100K concepts	Implicit sentiment, aspect-emotion pairing
ConceptNet	General commonsense graph	Millions of relations	Implicit aspect inference, event-valence reasoning
WordNet / SentiWordNet	Lexical database + sentiment	117K synsets	Synonym-based aspect generalisation
Domain ontology (custom)	Application-specific aspect hierarchy	Application-specific	Category assignment, ontology-aligned ASQP

8.2 LLMs for Zero-Shot and Few-Shot ABSA

Large language models (GPT-4, LLaMA 2, Mistral) applied to ABSA via prompting demonstrate strong zero-shot and few-shot performance on many ABSA sub-tasks, without any fine-tuning on ABSA-labelled data. This finding is consequential for practitioners: for many commercial ABSA applications, a well-prompted LLM may provide adequate performance without the expense and complexity of building and maintaining a labelled ABSA training pipeline.

The chain-of-thought prompting approach (Day 21, Section 4.2) is particularly effective for ABSA, where the structured reasoning chain naturally decomposes the multi-step problem: 'Step 1: Identify all entities or attributes being discussed. Step 2: For each entity/attribute, find the opinion words or expressions directed at it. Step 3: Classify the sentiment implied by each opinion expression as positive, negative, or neutral.' This decomposed prompting strategy guides the model through the same sub-tasks (ATE, then ASC, then ASTE) that pipeline systems implement as separate components, but within a single inference call.

```
# Chain-of-thought ABSA prompting (GPT-4 / LLaMA)
ABSA_COT_PROMPT = '''
Perform aspect-level sentiment analysis on the following review sentence.
Think step by step:
1. Identify all aspect terms (product features or service attributes mentioned).
2. For each aspect, identify the opinion expression (the words that describe it).
3. Classify the sentiment toward each aspect as: positive, negative, or neutral.
Output format: (aspect, opinion, sentiment) | (aspect, opinion, sentiment) ...

Review: {sentence}
'''

response = llm.generate(ABSA_COT_PROMPT.format(
    sentence='The battery life is disappointing but the camera is excellent.'
))
# Expected: (battery life, disappointing, negative) | (camera, excellent, positive)
```

8.3 Conversational ABSA

Standard ABSA operates on individual sentences or short passages in isolation. Conversational ABSA extends the task to multi-turn dialogue: a customer service conversation, a product discussion thread, or a therapy session transcript, where opinion expressions about aspects unfold across multiple speaker turns and where aspect references may be implicit (resolved through coreference), cumulative (aspect sentiment building across turns), or contradicted (a user expressing negative sentiment in one turn and moderating it in a later turn).

The challenges specific to conversational ABSA include: coreference resolution (identifying that 'it' in 'it worked perfectly' refers to the 'charging cable' mentioned two turns earlier), turn-level context modelling (the sentiment trajectory across turns may be more informative than any individual turn in isolation), and speaker role awareness (complaints from customers and responses from agents require different interpretation). Conversational ABSA is an active research area with several dedicated datasets (ABSA-MRC, DialogSA) and models emerging in 2023–2025, though it remains substantially less mature than sentence-level ABSA.

A practical illustration of conversational ABSA's value in production: in a contact centre conversation spanning eight turns, a customer first mentions that the delivery was fast (SERVICE#DELIVERY:

positive), then complains about incorrect items (PRODUCT#ACCURACY: negative), then expresses frustration about the return process (SERVICE#RETURNS: negative), and finally acknowledges that the replacement was sent promptly (SERVICE#DELIVERY: positive again). Sentence-level ABSA on any individual turn provides only partial insight; conversational ABSA tracking the full arc of the interaction surfaces the complete picture — and the turn at which each aspect's sentiment shifted — which is far more actionable for identifying systemic service failure points than any single-sentence ABSA reading.

```
# Conceptual conversational ABSA: maintaining aspect state across turns
conversation = [
    {'speaker': 'customer', 'text': 'The delivery was surprisingly fast!'},
    {'speaker': 'customer', 'text': 'But the wrong item was sent completely.'},
    {'speaker': 'agent',    'text': 'I am so sorry to hear that.'},
    {'speaker': 'customer', 'text': 'And the return process is incredibly complicated.'},
]
aspect_state = {}  # tracks aspect -> [polarity per turn]
for turn in conversation:
    if turn['speaker'] == 'customer':
        triplets = absa_model.extract(turn['text'])
        for aspect, opinion, polarity in triplets:
            aspect_state.setdefault(aspect, []).append(polarity)
# Final: {DELIVERY: ['pos'], PRODUCT_ACCURACY: ['neg'], RETURNS_PROCESS: ['neg']}
```

8.4 Open Problems in ABSA

Despite substantial recent progress, several ABSA challenges remain open problems where the gap between model and human performance is large enough to limit practical deployment:

- **Low-resource and zero-shot ABSA:** Most ABSA models require domain-specific labelled training data — which is expensive to collect for every new application domain. Zero-shot ABSA using large language models (Section 8.2) provides a practical path for new domains, but the gap between zero-shot LLM performance and full fine-tuned model performance remains 5–15 F1 points on most benchmarks. Closing this gap — through better prompting strategies, cross-domain transfer learning, or more efficient labelling pipelines — is a high-priority research direction.
- **ABSA robustness to adversarial inputs:** The ARTS benchmark (Section 2.2) demonstrated that current state-of-the-art models degrade substantially on adversarially constructed test cases that preserve the aspect-polarity label while changing surface form. Models that perform at 85–90% F1 on standard SemEval test sets can drop to 70–75% on ARTS, suggesting they are relying on superficial patterns (co-occurrence of aspect and nearby opinion words) rather than robust semantic understanding.
- **Temporal and causal aspect reasoning:** Standard ABSA treats each sentence as independent. Reviews frequently contain temporal or causal reasoning about aspects: 'The battery used to last all day; now it barely makes it to lunch' or 'The new software update has made the performance noticeably worse.' These temporal and causal sentiment patterns require reasoning beyond the single-sentence level and are outside the capability of current ABSA models.
- **Unified ABSA and emotion recognition:** Aspect-level sentiment (positive/negative/neutral) and aspect-directed emotion (frustrated about service, delighted with food) are closely related but typically modelled by separate systems. Unified models that jointly predict ABSA labels and fine-grained emotion labels would provide richer, more actionable customer insight but require more complex annotation and multi-task modelling — an underexplored direction with strong practical motivation.

8.5 Summary: ABSA and Emotion Recognition Landscape

Task / Approach	Current Best Method	Key Dataset	State-of-the-Art F1
ATE (Aspect Term Extraction)	BERT token classifier / joint	SemEval 2014 Rest.	~85% F1
ASC (Aspect Sentiment)	BERT-PT + aspect conditioning	SemEval 2014 Rest.	~85–88% Acc.
ASTE (Triplet Extraction)	Generative T5 / table-filling	SemEval 2015/2016	~70–74% F1
ASQP (Quad Prediction)	Generative T5 (GAS)	SemEval 2016	~55–60% F1
6-class Emotion	BERT fine-tuned	ISEAR, WASSA	~75–80% F1
27-class Emotion (GoEmotions)	BERT fine-tuned	GoEmotions	~46% macro-F1
Multimodal Sentiment	Cross-modal Transformer	CMU-MOSI	~84–86% Acc.

The future of AI is bright, and you can be part of it!